MOSTLY·AI

Data – The New Lingua Franca

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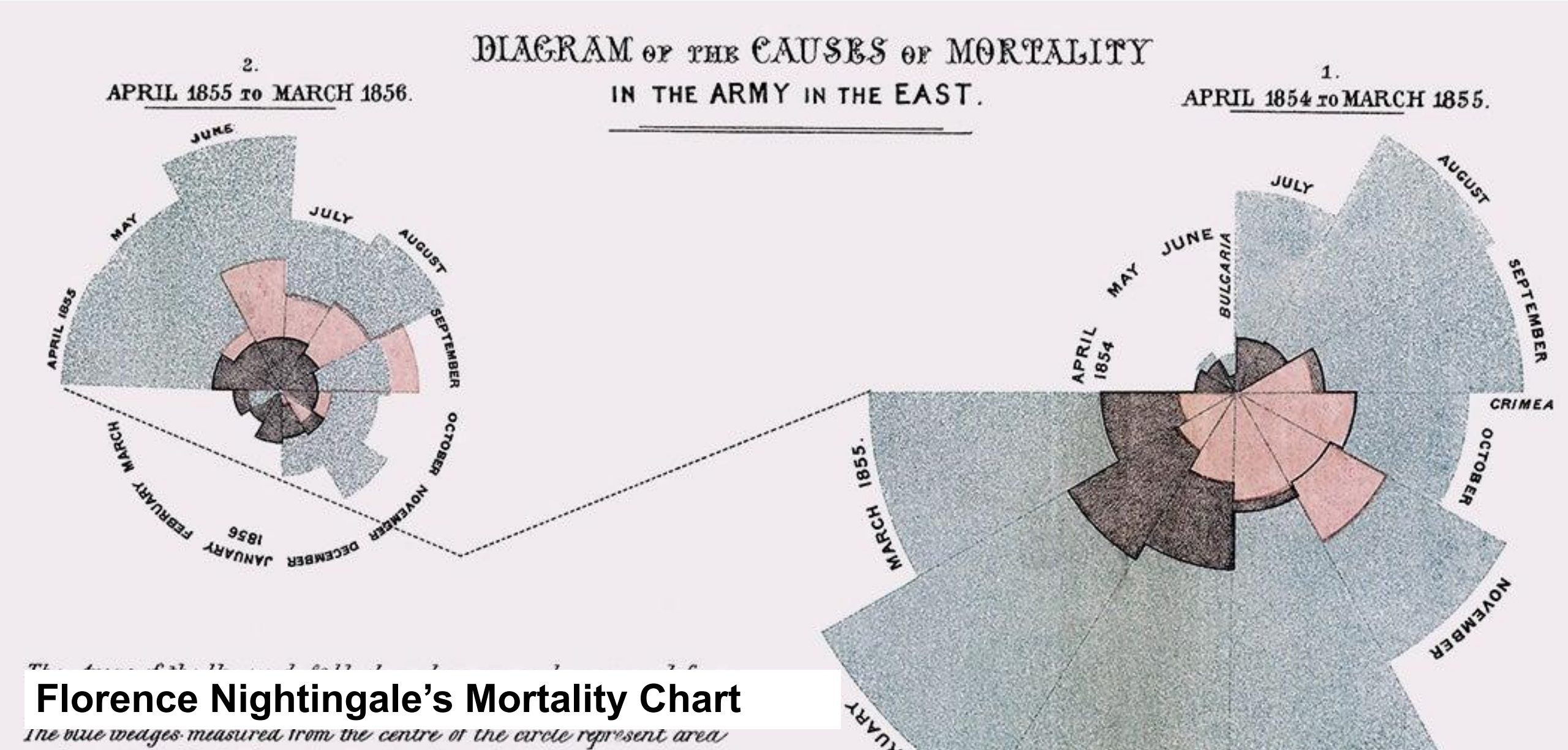
May 2023

3 Megatrends

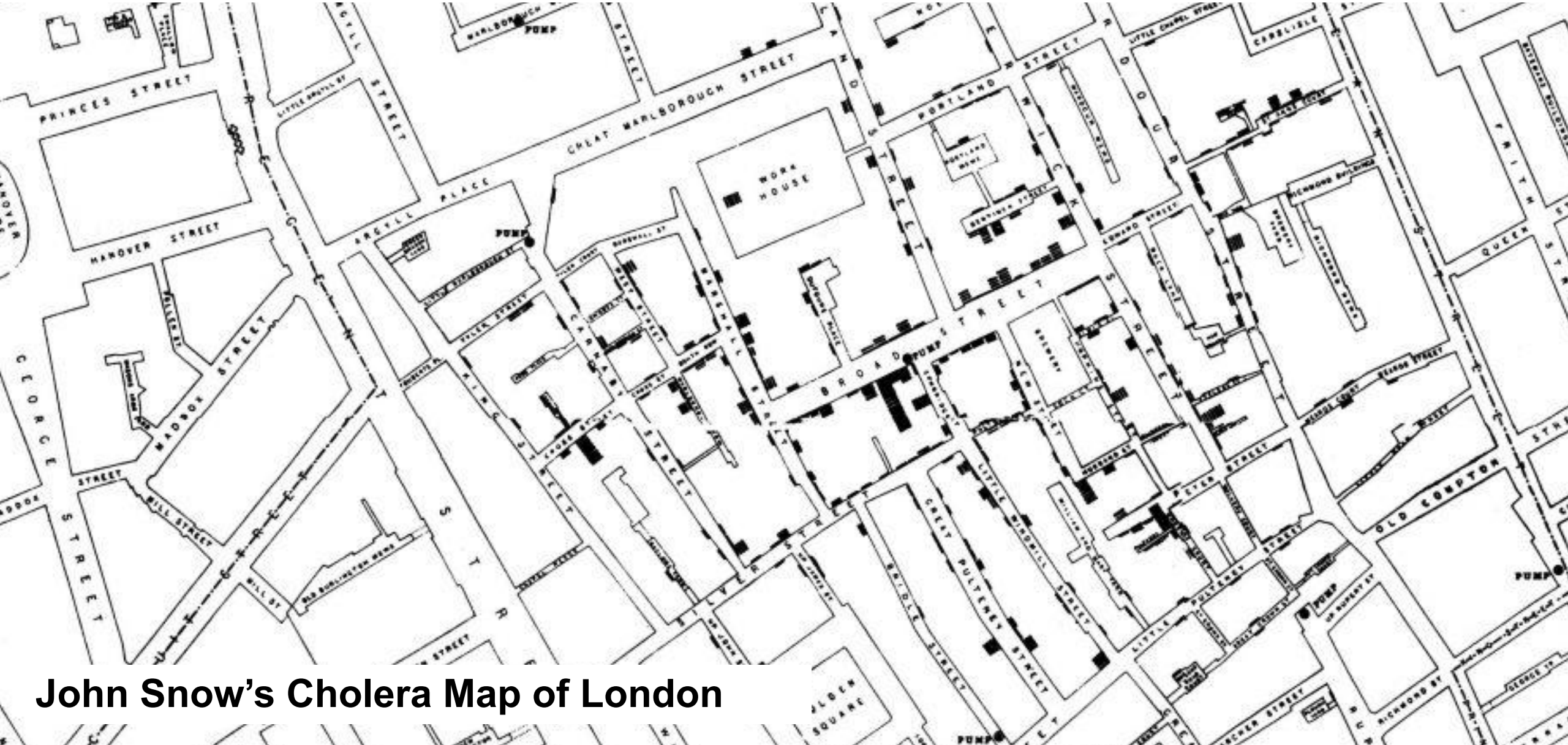
- The world is interdependent **Collaboration**
- Data volumes are growing **Data Protection**
- The algorithms win **AI Regulation**

Data – The ~~New~~ Old Lingua Franca



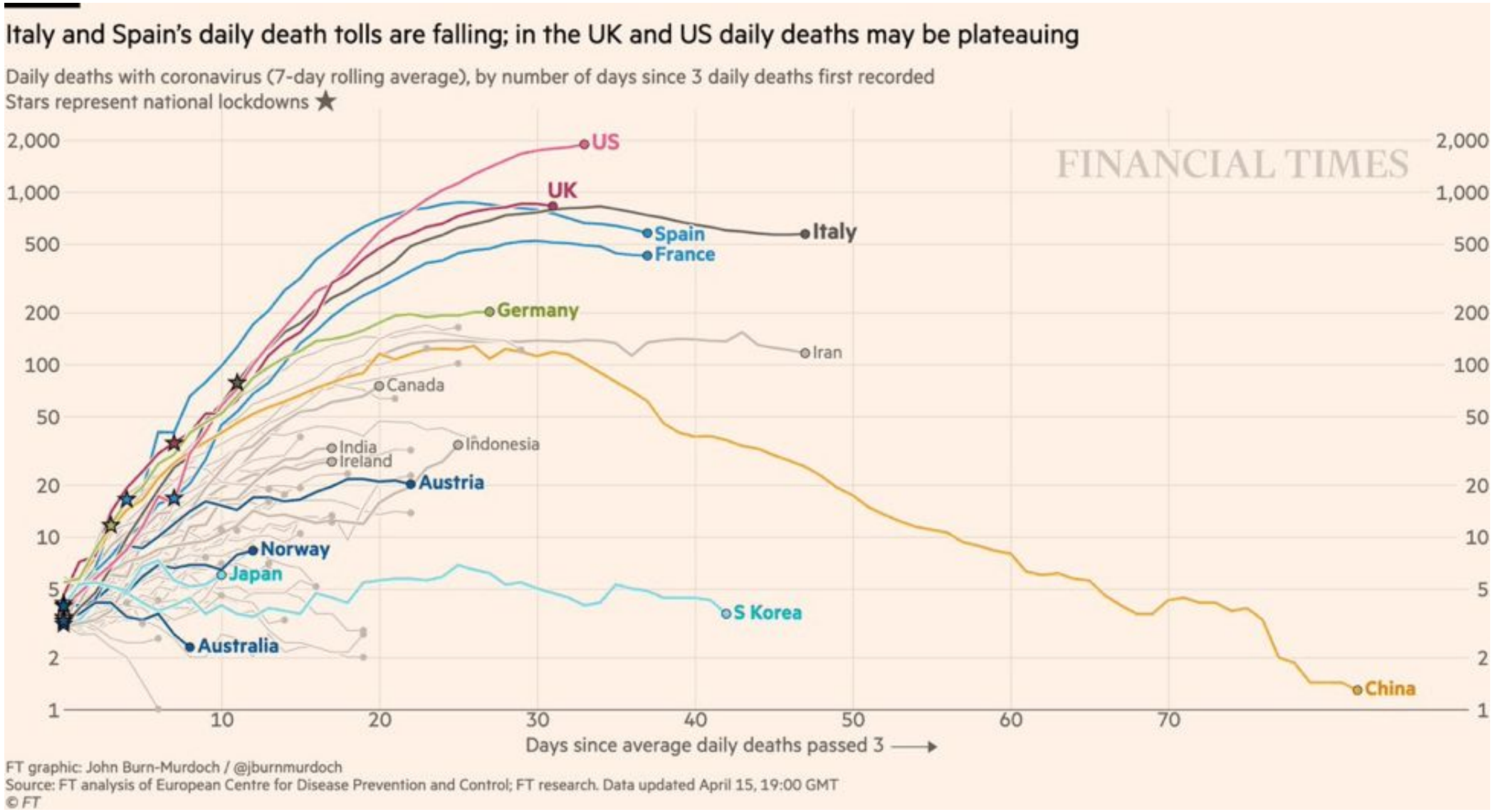
Florence Nightingale's Mortality Chart

Data – The ~~New~~ Old Lingua Franca

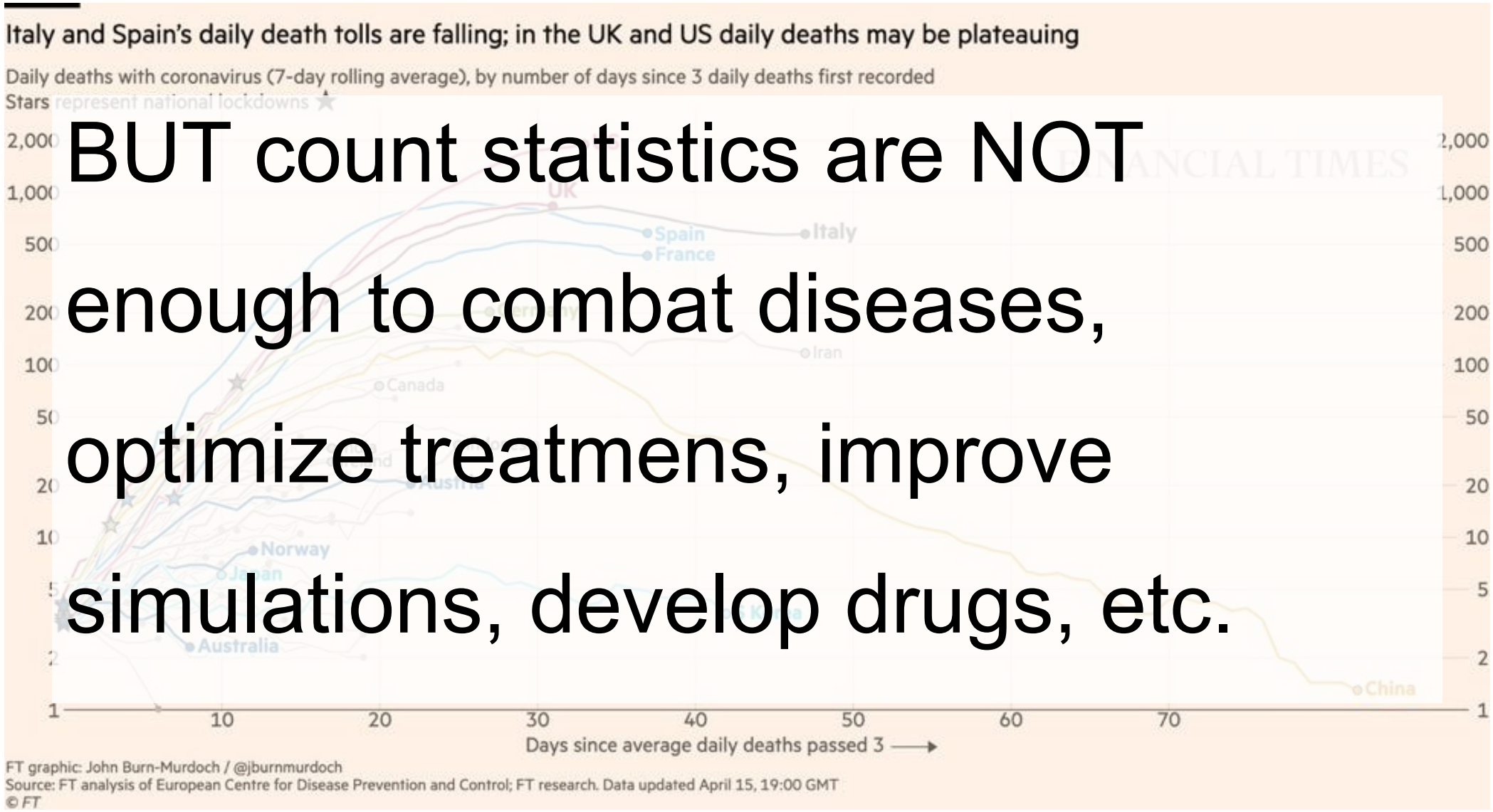


John Snow's Cholera Map of London

Data – The ~~New~~ Old Lingua Franca



Data – The ~~New~~ Old Lingua Franca



BUT count statistics are NOT enough to combat diseases, optimize treatmens, improve simulations, develop drugs, etc.

Micro Data – The New Lingua Franca

43 years, Male
Weight: 98kg
Height: 1m76cm
Smoker: ~5/day
Ethnicity: […]
Profession: […]
Location: […]
Pre-existing conditions: […]
Continuous medication: […]
Treatment history: [...]
Measures history: […]
Symptoms history: […]
Social activity: […]

Thousands of Data Points for Millions of Patients

Daten – Die Neue Weltsprache

43 years, Male

Weight: 98kg
Height: 1m76cm
Smoker: ~5/day
Ethnicity: […]
Profession: […]
Location: […]
Pre-existing conditions: […]
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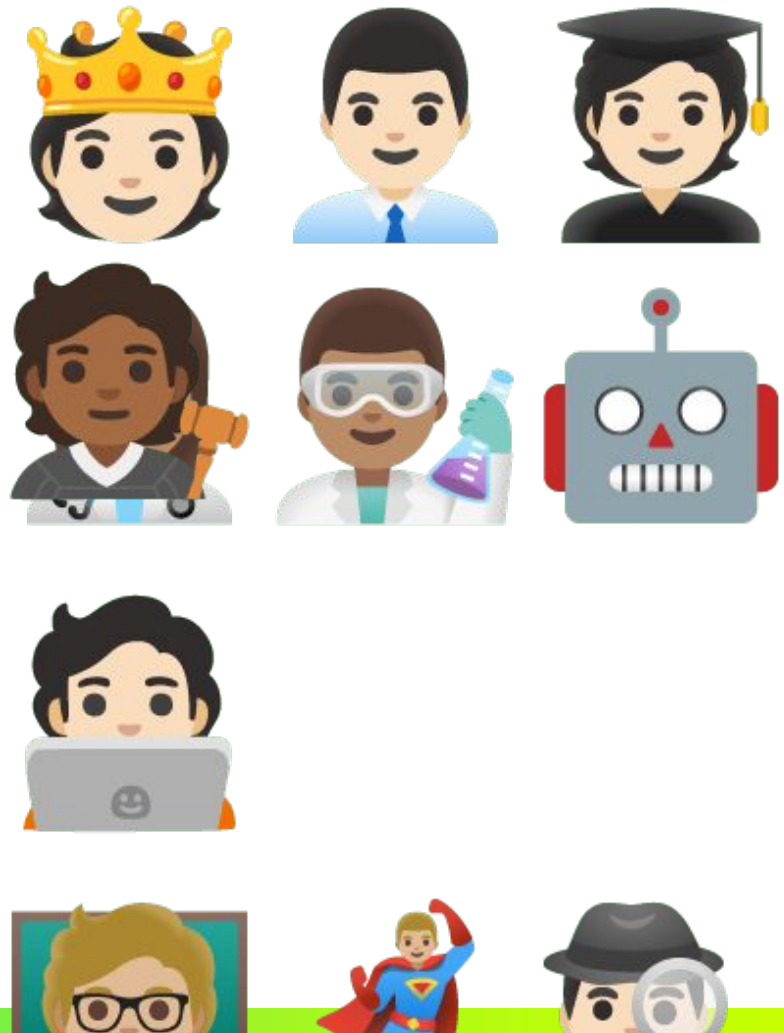
Social activity: […]

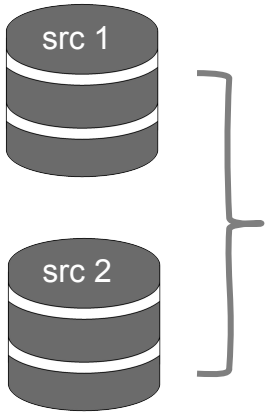
For millions of Data Consumers



Data as The Foundation for a Healthy Democracy

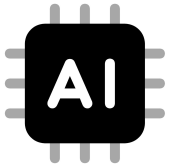
- Data for Research
- Data for Education
- Data for Economy
- Data for Innovation
- Data for Policy Making
- Data for Public Discours
- Data for Developing Algorithms
- Data for Validating Algorithms
- Data for Regulating Algorithms

Statistically representative, granular level synthetic Data

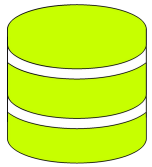


Actual Data

privacy-restricted

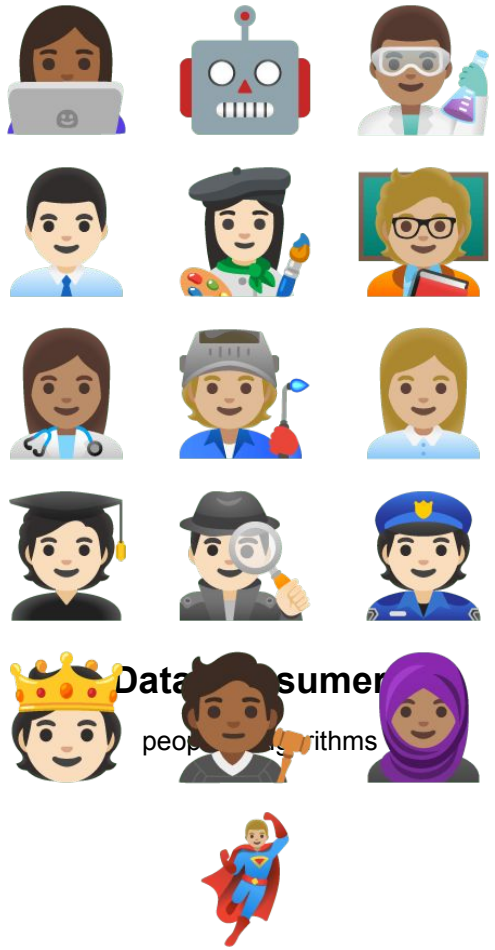


MOSTLY·AI



Synthetic Data

realistic
representative
truly anonymous
granular level



Data Consumer

people, algorithms

🇦🇹 Weltmarktführer MADE IN AUSTRIA 🇦🇹

Statistically representative, granular level synthetic Data

87 years, Female
Weight: 76kg
Height: 1m54cm
Smoker: None
Ethnicity: […]
Profession: […]
Location: […]
Pre-existing conditions: […]
Continuous medication: […]
Treatment history: [...]
Measures history: […]
Symptoms history: […]
Social activity: […]

Thousands of Data Points for Millions of Virtual Patients.

Same Structur,
Same Statistics,
but no real people.

Privacy is never at risk.

Key Question: How accurate is Synthetic Data really?

https://publications.jrc.ec.europa.eu/repository/handle/JRC128595

- 89-page Technical Report by **Joint Research Centers**
- Synthesis of **2 Millionen Cancerpatients**
- End-to-end synthesis on top of **MOSTLY AI** platform
- Use Case: **Cross-Border Research**

Key Question: How accurate is Synthetic Data really?

We can check the consistence of the synthetic data by plotting the same graph but with a different colour scale for the deceased patients:

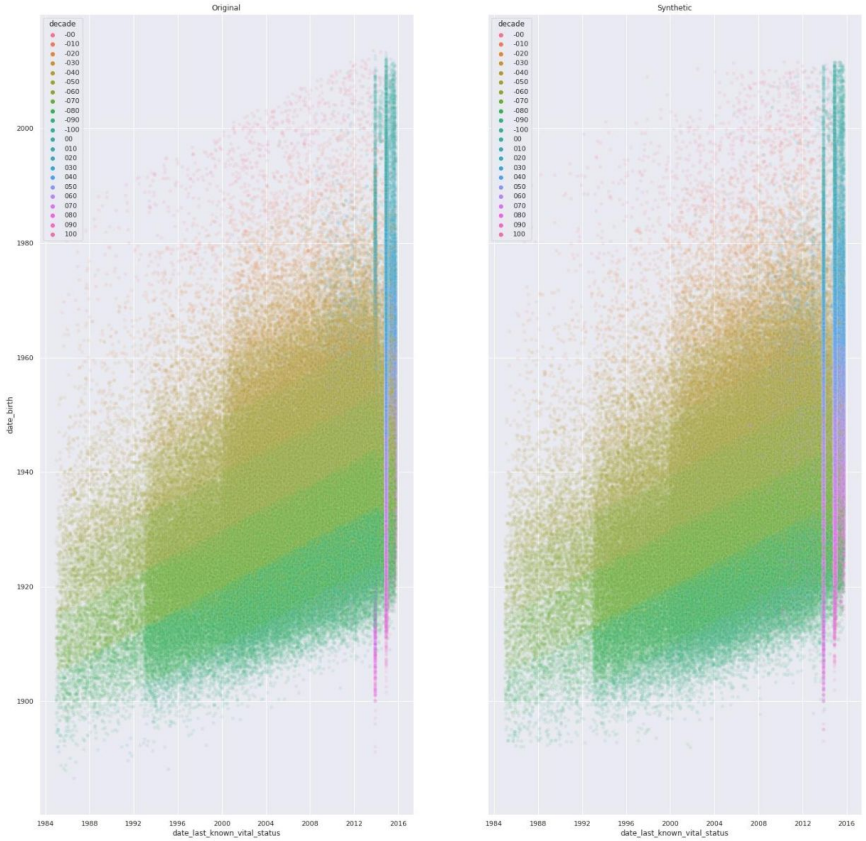


Figure 37 Date of birth, last known vital status, incidence, in original dataset (left) and synthetic dataset (right) wit deceased patients in a different colour scale.

Another way to demonstrate consistence was elementary descriptive statistics, here for age distribution by cancer groups:

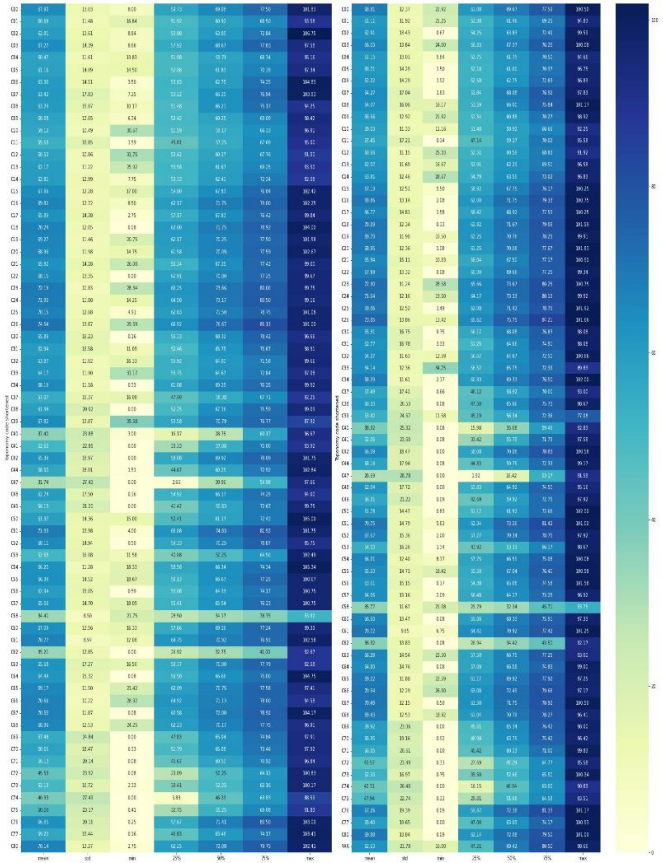
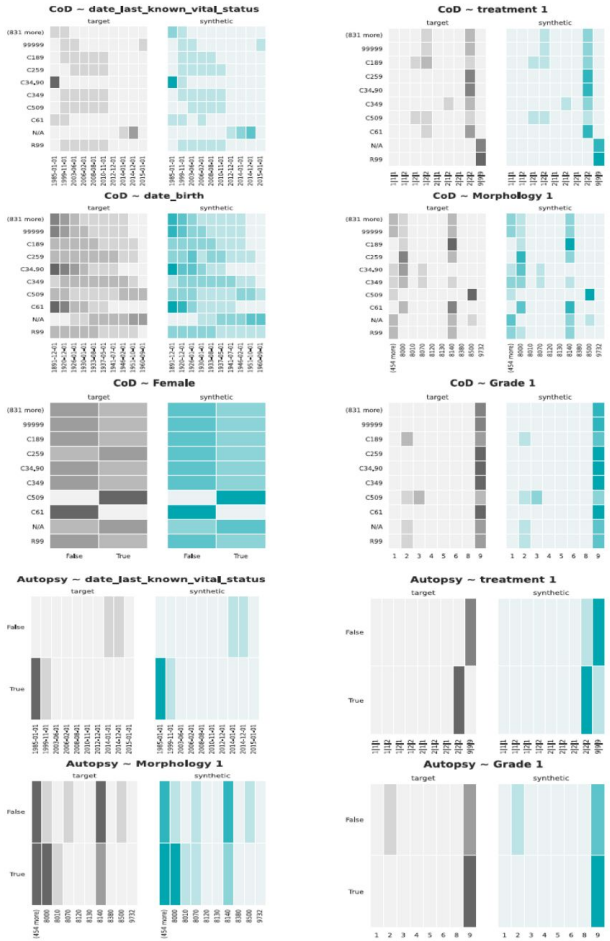


Figure 38 Age distribution by cancer groups, in original (left) and synthetic (right) dataset.



Key Question: How accurate is Synthetic Data really?

Synthetic data will be the key enabler for AI

- "[Synthetic data] can become the unifying bridge between policy support and computational models, by unlocking the potential of data hidden in silos; thus **becoming the key enabler of AI in business and policy applications in Europe**."
- "Resulting data not only can be shared freely, but also can help rebalance under-represented classes in research studies via oversampling, making it the **perfect input into machine learning and AI models**"

It is very cost efficient to use synthetic data for privacy protection

- "**Synthetic data change[s] everything** from privacy to governance"
- "**Synthetic data eliminates the bureaucratic burden** associated with gaining access to sensitive data"
- "among the privacy-preservation technique studies analysed, data synthesis gave the **best price/effort ratio**."

Synthetic data is a mature privacy enhancing technology (PET) that is ready to be deployed

- "Synthetic data have proven great potential and are **the go-to methods ready to be deployed in real-life scenarios**. Policy applications can now be researched and developed with little risk involved."
- "Commercial solutions still beat the available research and open source solutions by a **huge margin** at the time of writing."
- "A robust commercial solution for hierarchical data synthesis was offered by the company **MOSTLY.AI** to test the synthesis quality. **The results**, available in the accompanying archive both as a PDF report and CSV files with the data correlations, **are impressive**."

Synthetic data improves fairness, diversity and inclusion

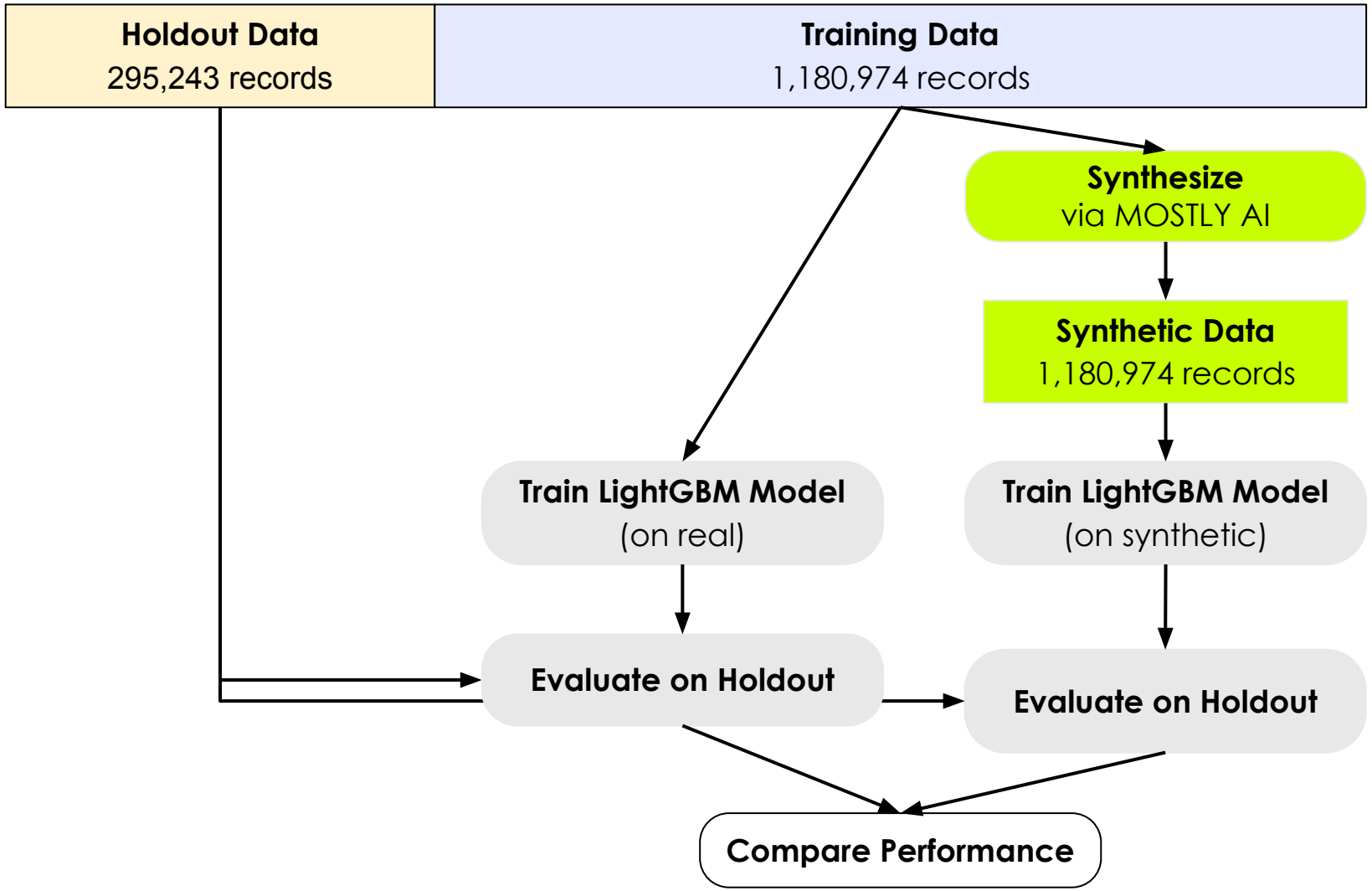
- "...policy makers will have the opportunity to create radically new policy instruments based on more detailed **less biased population data.**"
- "policy support can **embrace diversity** by stopping averaging out, and thus marginalisation of under-represented minorities by shifting away from aggregate-level statistics; capture the full diversity within the population, and start coping with its complexity, rather than continue ignoring it."

Synthetic data facilitates open data, data democratization, and data literacy

- "**Data democratization** – a new type of Open Data would emerge from the shift towards the synthetic data"
- "synthetic data in policy making process will enable **involvement of much broader communities** [..] same goes for **algorithmic governance**"
- "**Data literacy** is today as what was computer literacy in 1990s. Synthetic data have potential to accelerate their learning path towards data-driven decision & policy making by making available data closer to the people's perspective."

"Synthetic data will be the key enabler for AI in Europe."

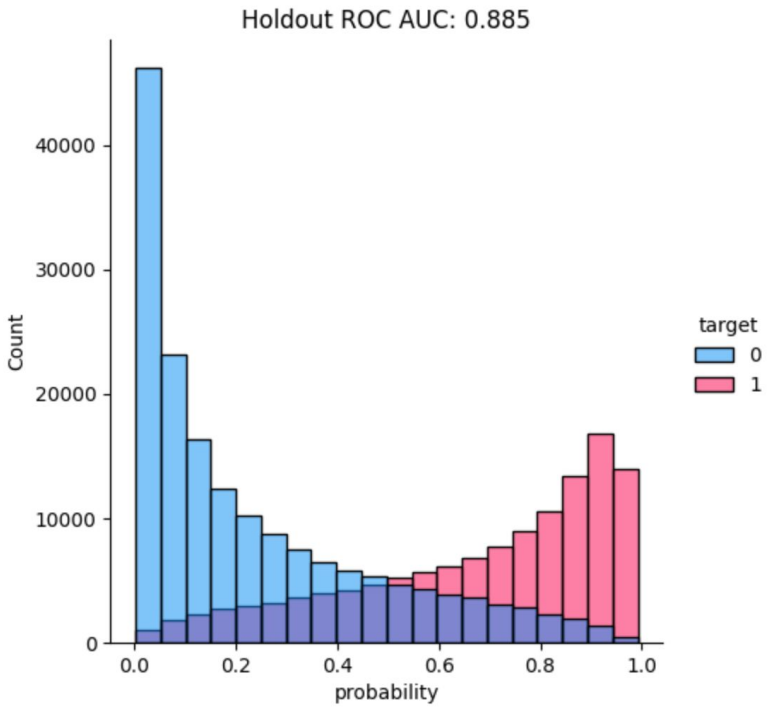
Synthetic Data for AI Training



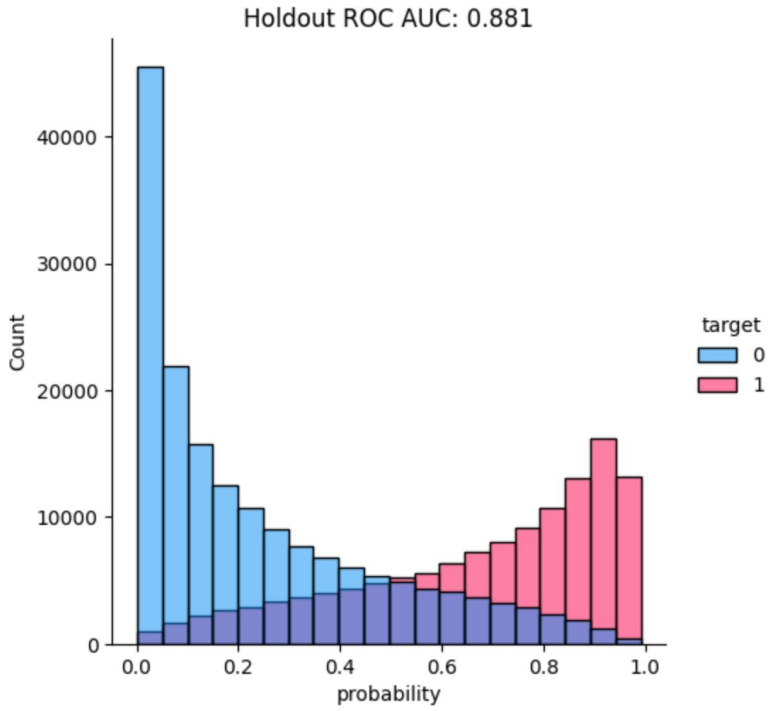
Synthetic Data for AI Training

Identical Performance on Actual Data

Train-Real-Score-Real



Train-Synthetic-Score-Real

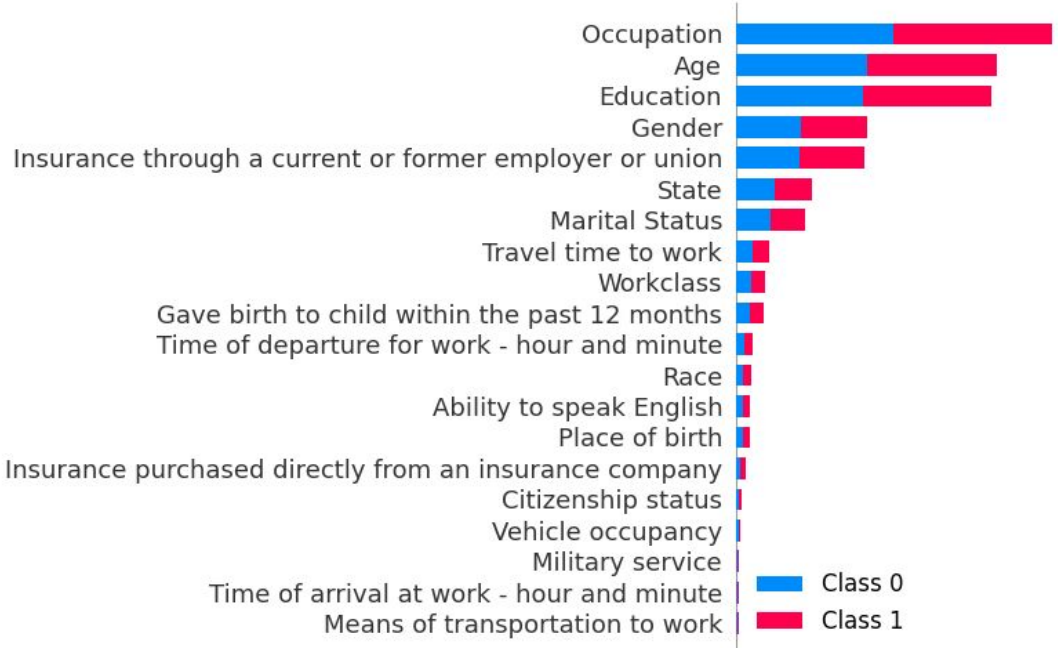


When assessing the model performance on an actual holdout dataset, we see near identical predictive accuracy, as well as distribution of assigned probability scores.

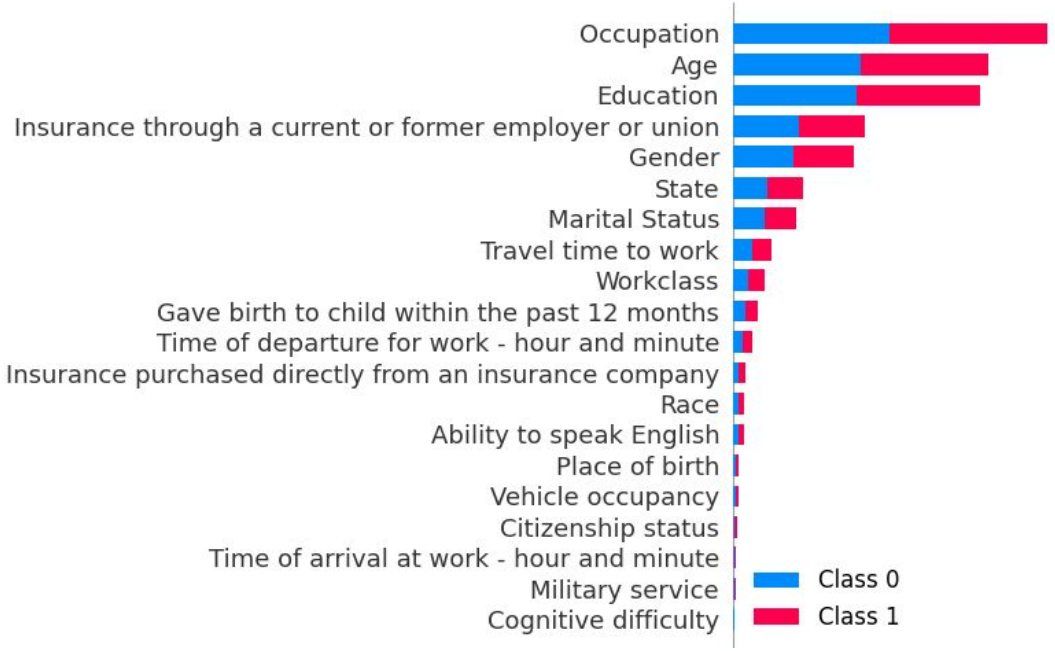
Synthetic Data for AI Validation

Identical Findings for Driver Analysis

Train-Real-Explain-Real



Train-Synthetic-Explain-Synthetic



Synthetic Data for AI Explainability

Identical Findings for Driver Decomposition

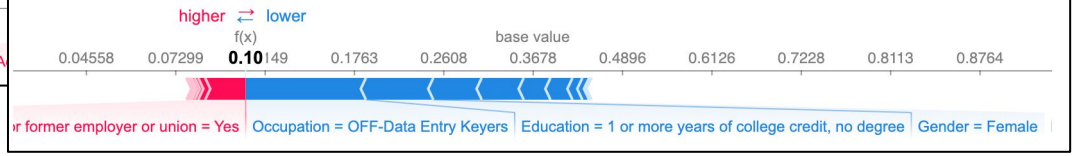
Train-Real-Explain-Real

```
Customer:    100
Target:      1
Probability: 52.25%

Customer:    101
Target:      1
Probability: 21.04%

Customer:    100
Target:      0
Probability: 10.46%

##### Top 8 Attributes #####
Occupation                                                    OFF-Data Entry Keyers
Education                                 1 or more years of college credit, no degree
Gender                                                                       Female
Marital Status                                     Never married or under 15 years old
Insurance through a current or former employer or union                          Yes
Gave birth to child within the past 12 months                                     No
State                                                                        Ohio/OH
Vehicle occupancy                                                                2.0
Name: 100, dtype: object
```



Train-Synthetic-Explain-Synthetic

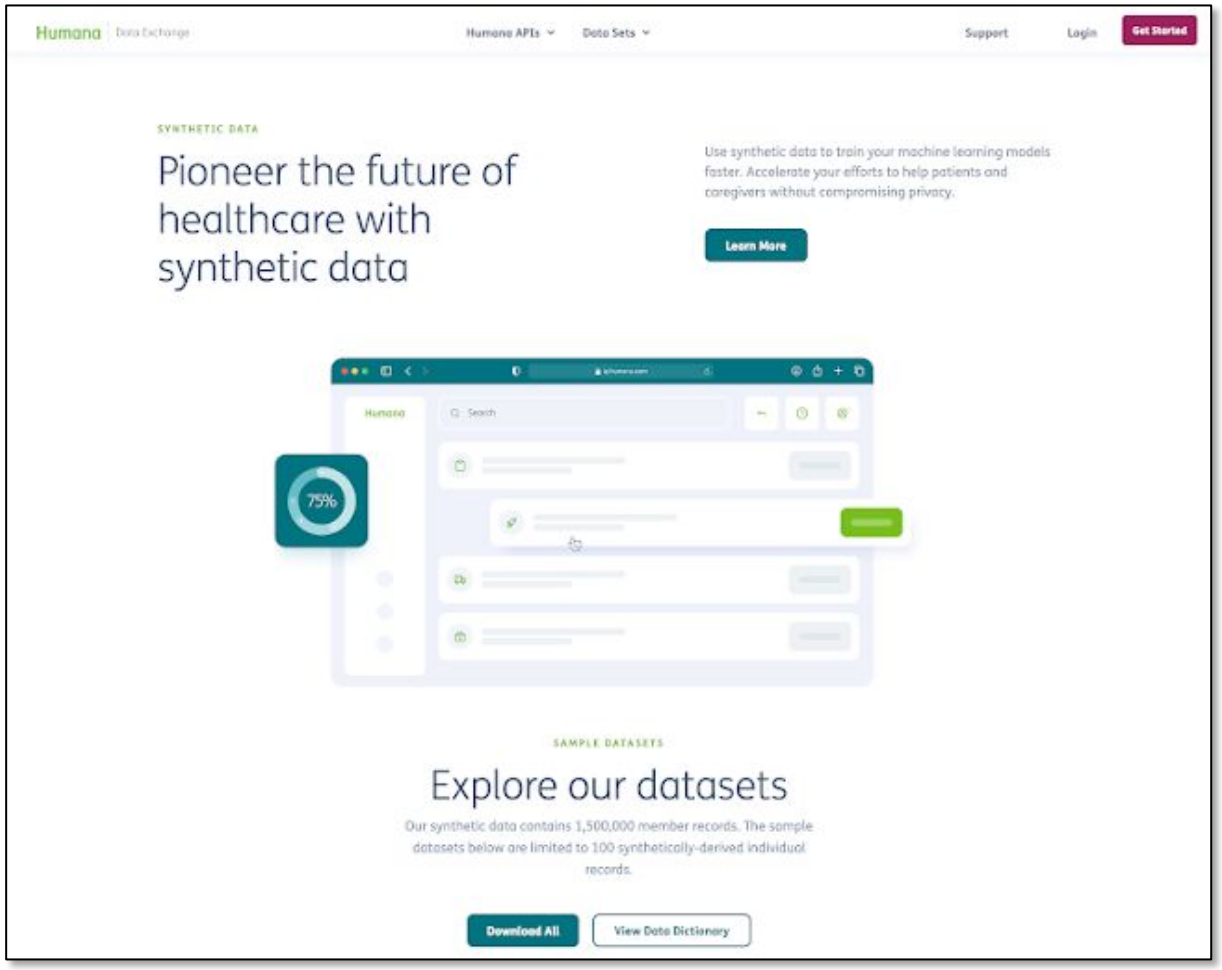
```
Customer:    100
Target:      1
Probability: 52.25%

Customer:    101
Target:      1
Probability: 21.04%

Customer:    102
Target:      1
Probability: 85.11%

##### Top 8 Attributes #####
Education                                                        Master's degree
Age                                                                         50.0
Occupation                                   MGR-Education And Childcare Administrators
Gender                                                                     Female
Insurance through a current or former employer or union                       Yes
Travel time to work                                                         147.0
State                                                              New Hampshire/NH
Marital Status                                                            Married
Name: 102, dtype: object
```

Healthcare Data Exchange – powered by MOSTLY AI



- https://developers.humana.com/syntheticdata

BUT…

Data Protection is not only a technical barrier, but also a convenient pretense to defend existing **Data Monopolies**, as well as avoid transparency.

Call to Action

Let’s work together to combat data monopolies, foster data literacy, as well as establish an open data democracy. All of these are needed to be ready for the upcoming challenges in health care.